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Parallel Pricing Two-Asset American Options by Projected SOR: A linear complementarity formulation on a distributed five-point stencil

Abstract

We price a two-asset American put under uncorrelated geometric Brownian motion by solving the Black-Scholes partial differential equation as a sequence of linear complementarity problems, one per time step, each discretized on a five-point stencil and solved by projected successive over-relaxation with red-black ordering. The solver is parallelized by two-dimensional domain decomposition over an MPI Cartesian communicator. On three 28-core Intel Xeon E5-2683 v3 nodes of SeaWulf we measure a speedup of 45.65 on 64 cores at a 1024×1024 mesh, corresponding to 71.3% efficiency and a Karp-Flatt metric of 0.0064. Three independent correctness criteria are satisfied. Results are bit-identical across 1 to 64 cores spanning three physical nodes; disabling the projection recovers the closed-form European price to within 0.32%; and the deep-in-the-money early-exercise premium reproduces the analytic value $2K(1-e^{-rT}) = 9.7541151$ to seven significant figures. We find that parallel efficiency at fixed core count rises monotonically with mesh size, from 10.9% to 71.3% at $P = 64$, and attribute this to the surface-to-volume ratio of the per-rank subdomain, which we quantify directly. We further observe that, unlike the matrix multiplication of Project 2, a partitioned-data stencil solver has a per-socket cache footprint independent of P, which explains the absence of any L3 capacity cliff in the present measurements and predicts its onset near $N \approx 1480$ on this hardware.

1 Project Description

The difference between a European and an American option is a single decision, and that decision governs everything about how the two are computed. A European contract admits no choice before expiry. Its value is a conditional expectation under the risk-neutral measure,

$$V(S,t) = E^{Q}[\, e^{-r(T-t)}\, \Phi(S_T)\,]$$

and it satisfies the Black-Scholes equation [1] with equality throughout the domain. An American contract may be exercised at any time up to expiry, so its value is the supremum of that expectation over all admissible stopping times. The holder's freedom partitions the domain into a continuation region, in which holding is optimal and the equation holds exactly, and an exercise region, in which the value equals the payoff and the equation degrades to an inequality. The interface between them is a free boundary whose position is not known in advance and which migrates as time to maturity decreases. No closed-form solution exists. For two uncorrelated assets the governing equation carries no cross-derivative term and reads

$$V_t + \tfrac{1}{2}\sigma_1^2 S_1^2 V_{S_1S_1} + \tfrac{1}{2}\sigma_2^2 S_2^2 V_{S_2S_2} + rS_1V_{S_1} + rS_2V_{S_2} - rV = 0$$

The absence of the mixed partial is consequential rather than cosmetic: it is precisely what permits the five-point stencil of Problem 3.4 to be carried over unchanged. A correlated pair would require a nine-point stencil and a correspondingly wider halo.

1.1 Choice of Payoff

We take the separable payoff

$$\Phi(S_1,S_2) = \max(K_1 - S_1,\, 0) + \max(K_2 - S_2,\, 0)$$

Two deliberate choices are embedded here, and both serve verification. First, the payoff is separable, so by linearity of expectation the European value is exactly the sum of two one-dimensional Black-Scholes puts. The European configuration therefore has an analytic answer against which the discretisation can be measured, which is what the assignment asks for, and the exactness survives into the Dirichlet boundary data.

Second, the contracts are puts rather than calls. On a non-dividend-paying asset an American call is never exercised early when interest rates are positive - selling is more profitable. Were calls used, the continuation region would fill the entire domain, the projection step would never bind, and the linear complementarity problem would collapse to an ordinary linear system with nothing to demonstrate. Puts possess a genuine free boundary and so exercise the machinery the problem is about.

1.2 Parameters

Quantity	Symbol	Value
Strikes	K1, K2	100.0, 100.0
Volatilities	sigma1, sigma2	0.30, 0.25
Risk-free rate	r	0.05
Maturity	T	1.0 year
Truncated domain	[0,Smax]^2	[0, 300]^2
Interior mesh	M x M	128, 256, 512, 1024
Time steps (backward Euler)	nt	50
SOR tolerance	tol	1.0e-7
Core counts	P	1, 2, 4, 16, 64

Table 1. Model and discretisation parameters. The relaxation factor is not tuned by hand but set to the theoretical optimum for a five-point stencil, $\omega = 2/(1 + \sin(\pi/(M+1)))$ [6], giving 1.9525 at M = 128 and 1.9939 at M = 1024.

2 Program Structure

2.1 From a parabolic PDE to a sequence of linear complementarity problems

Substituting the time to maturity $\tau = T - t$ reverses the direction of integration, so that the terminal payoff becomes an initial condition and the solution marches forward in τ. Discretizing that derivative by backward Euler produces, at each step,

$$(I - \Delta\tau\, L)\, V^{n+1} = V^{n}$$

in which L carries exactly the five-point stencil of the Poisson operator in Problem 3.4. In this precise sense Problem 3.5 is Problem 3.4 executed fifty times in succession, with two structural differences. The identity contributes to the diagonal, so that the central coefficient is $1 + \Delta\tau(2a_i + 2c_j + r)$ rather than a constant 4; and the coefficients vary from node to node, because a and c scale with S^2. Writing $a_i = \tfrac{1}{2}\sigma_1^{\,2} S_{1,i}^{\,2}/h_1^{\,2}$ and $b_i = rS_{1,i}/(2h_1)$, the five weights are

$$w_W = \Delta\tau(a-b),\quad w_E = \Delta\tau(a+b),\quad w_S = \Delta\tau(c-d),\quad w_N = \Delta\tau(c+d),\quad w_C = 1 + \Delta\tau(2a+2c+r)$$

For the American contract the equality is replaced by three simultaneous conditions. Writing $A = I - \Delta\tau L$, the discrete problem at each step is the linear complementarity problem

$$V - \Phi \geq 0,\quad AV - V^{n} \geq 0,\quad (V - \Phi)^{T}(AV - V^{n}) = 0$$

The first condition is a no-arbitrage statement: the option can never be worth less than immediate exercise. The third, the complementarity condition, asserts that at every node one of the first two holds with equality. The free boundary is consequently never located explicitly. It is revealed, as the set across which the active condition changes, which is the principal advantage of this formulation over front-tracking alternatives [3, 4].

2.2 Projected SOR and why it converges here

The solver is ordinary SOR with a projection applied after each update, which in the code is a single line:

```
vn = V[q] + omega * (gs - V[q]);
if (american && vn < Phi[q]) vn = Phi[q];
```

That second line is the entire American feature. It is a projection onto the convex set $\{V \geq \Phi\}$. Deleting it recovers a European option, which is exactly what the -e flag does and what makes the validation of Section 3 possible. Convergence is not assumed. The convection terms $rS \cdot V_S$ could in principle render an off-diagonal weight positive, destroying the M-matrix property on which convergence of projected SOR rests. The condition $w_W \geq 0$ requires $\frac{1}{2}\sigma^2 S^2/h^2 \geq rS/(2h)$, that is $i \geq r/\sigma^2$. With $r = 0.05$ this threshold is 0.56 for the first asset and 0.80 for the second, so it is satisfied at every interior node $i \geq 1$. The matrix is therefore a strictly diagonally dominant M-matrix and projected SOR converges to the unique solution of the complementarity problem [5, 7].

2.3 Red-black ordering and invariance under decomposition

Lexicographic SOR is inherently sequential: each update consumes values produced earlier in the same sweep. Red-black ordering removes that dependence. Because the five-point stencil connects a node only to neighbours of the opposite parity, every node of one colour may be updated simultaneously without reference to any other node of the same color.

The coloring is assigned from global indices, not local ones. This is the decisive implementation detail. It guarantees that the sequence of iterates is a property of the mathematics alone and not of how the mesh happens to be divided, so that one core and sixty-four cores traverse an identical trajectory. Section 3.2 confirms this to ten significant figures and it is what licenses the speedup figures of Section 4: the comparison is of identical work, differently distributed, rather than of two different computations.

2.4 Parallel Decomposition

The mesh is divided into rectangular blocks over a two-dimensional Cartesian communicator, the factorization supplied by MPI_Dims_create, which yields 2×1 at P = 2, 2×2 at P = 4, 4×4 at P = 16 and 8×8 at P = 64. Each rank stores its block surrounded by a one-cell halo. In the S1 direction a halo row is contiguous in memory and is exchanged directly; in the S2 direction the halo is strided and is packed into a contiguous buffer first. All eight transfers are posted as non-blocking MPI_Isend and MPI_Irecv operations and completed by a single MPI_Waitall. Edge ranks address MPI_PROC_NULL, so no special-casing of boundaries is required. The build log records a `-Wstringop-overflow` diagnostic against `MPI_Waitall`; this is a known false positive arising from the zero-length array declaration of `MPI_STATUSES_IGNORE` in the mvapich2 headers, and the code compiles and runs correctly.

A halo exchange precedes each color, giving two exchanges per sweep, and one MPI_Allreduce of the maximum correction per sweep provides the convergence test. Wall time is taken from the slowest rank, identified by an MPI_MAXLOC reduction and broadcast, so that the reported communication share belongs to the same rank as the reported runtime. Section 4.4 discusses the limitations of this choice frankly. A further consequence of the formulation deserves emphasis. The projection max(V, Φ) is pointwise. It consults no neighbor, crosses no subdomain boundary and adds no message. The feature that makes the American problem analytically intractable is therefore, in parallel, entirely free.

3 Results and Verification

Three independent checks are applied. Each isolates a different part of the solver, and the third is analytic.

3.1 The European limit against the closed form

M	numeric V(K,K)	exact V(K,K)	abs. error	rel. error	max error
128	16.73965147	16.81313862	7.349e-02	0.437%	7.401e-02
256	16.46899245	16.52405354	5.506e-02	0.333%	5.575e-02
512	16.75769549	16.81313862	5.544e-02	0.330%	5.643e-02
1024	16.68593327	16.74026209	5.433e-02	0.325%	5.528e-02

Table 2. Disabling the projection must reproduce the sum of two Black-Scholes puts. Relative error falls from 0.437% to 0.325%. The residual floor is not a defect of the spatial discretisation: instrumenting the location of the maximum error places it at the strike, where the terminal payoff has a kink, which is the classical behaviour of finite differences applied to a non-smooth initial condition. Rannacher start-up [8] would reduce it further.

3.2 Independence of the core count

P	grid	SOR iterations	$\|V\|_2$	V(K,K)
1	1x1	27928	1.3489710295e+04	16.9934552067
2	2x1	27928	1.3489710295e+04	16.9934552067
4	2x2	27928	1.3489710295e+04	16.9934552067
16	4x4	27928	1.3489710295e+04	16.9934552067
64	8x8	27928	1.3489710295e+04	16.9934552067

Table 3. The American solution at M = 256 as the decomposition is varied from a single core to sixty-four across three physical nodes. The iteration count, the solution norm to eleven digits and the price to ten digits are invariant. This is the consequence of globally indexed red-black ordering described in Section 2.3, and it is a stronger statement than agreement to a tolerance.

3.3 The early-exercise premium, and an analytic check

M	American	European (exact)	premium	% of European
128	17.27740606	16.81313862	0.464267	2.761%
256	16.99345521	16.52405354	0.469402	2.841%
512	17.29720171	16.81313862	0.484063	2.879%
1024	17.22198281	16.74026209	0.481721	2.878%

Table 4. Early-exercise premium at the money. The premium converges to roughly 2.88% of the European value.

An exact test of the projection:

Every American run reports a maximum deviation from the European price of 9.754115, attained at S1 = S2 ≈ 0.6, deep in the money. That point admits an analytic answer. As S → 0 the American put is worth its strike immediately, while the European holder must wait and so holds only the discounted strike; the gap for the pair is $2K(1 - e^{-rT}) = 9.7541151$. *The computed value agrees to seven significant figures. This is a sharper instrument than the at-the-money comparison, because it isolates the projection step alone at a location where the free boundary is unambiguous. Interpreted financially, the quantity measured is the interest earned on the strike over the remaining life, which is precisely what early exercise purchases.*

At the money the premium is far smaller, near 2.88%, because that region lies close to the free boundary itself, where the decision to exercise is marginal and the two regimes nearly coincide. The contrast between the two figures is the economic content of the calculation.

4 Analysis of Program Performance

M	P	grid	time (s)	comm (s)	comm %	speedup	efficiency	Karp-Flatt
128	1	1x1	1.914	0.075	3.9%	1.00	1.000	—
128	2	2x1	1.078	0.067	6.2%	1.78	0.888	0.1261
128	4	2x2	0.649	0.167	25.7%	2.95	0.737	0.1187
128	16	4x4	0.272	0.104	38.5%	7.05	0.440	0.0847
128	64	8x8	0.274	0.238	86.6%	6.98	0.109	0.1297
256	1	1x1	12.540	0.188	1.5%	1.00	1.000	—
256	2	2x1	6.635	0.211	3.2%	1.89	0.945	0.0581
256	4	2x2	3.817	0.217	5.7%	3.29	0.821	0.0725
256	16	4x4	1.268	0.247	19.5%	9.89	0.618	0.0412
256	64	8x8	0.747	0.468	62.7%	16.79	0.262	0.0446
512	1	1x1	106.552	0.702	0.7%	1.00	1.000	—
512	2	2x1	53.397	1.047	2.0%	2.00	0.998	0.0023
512	4	2x2	27.671	0.706	2.6%	3.85	0.963	0.0129
512	16	4x4	8.050	0.833	10.3%	13.24	0.827	0.0139
512	64	8x8	3.025	1.168	38.6%	35.22	0.550	0.0130
1024	1	1x1	824.617	2.575	0.3%	1.00	1.000	—
1024	2	2x1	417.954	2.412	0.6%	1.97	0.986	0.0137
1024	4	2x2	234.357	7.150	3.1%	3.52	0.880	0.0456
1024	16	4x4	59.151	11.888	20.1%	13.94	0.871	0.0098
1024	64	8x8	18.065	3.935	21.8%	45.65	0.713	0.0064

Table 5. Complete performance sweep, American contract. S = T(1)/T(P), E = S/P, and the Karp-Flatt metric e = (1/S − 1/P)/(1 − 1/P) [11] estimates the experimentally determined serial fraction.

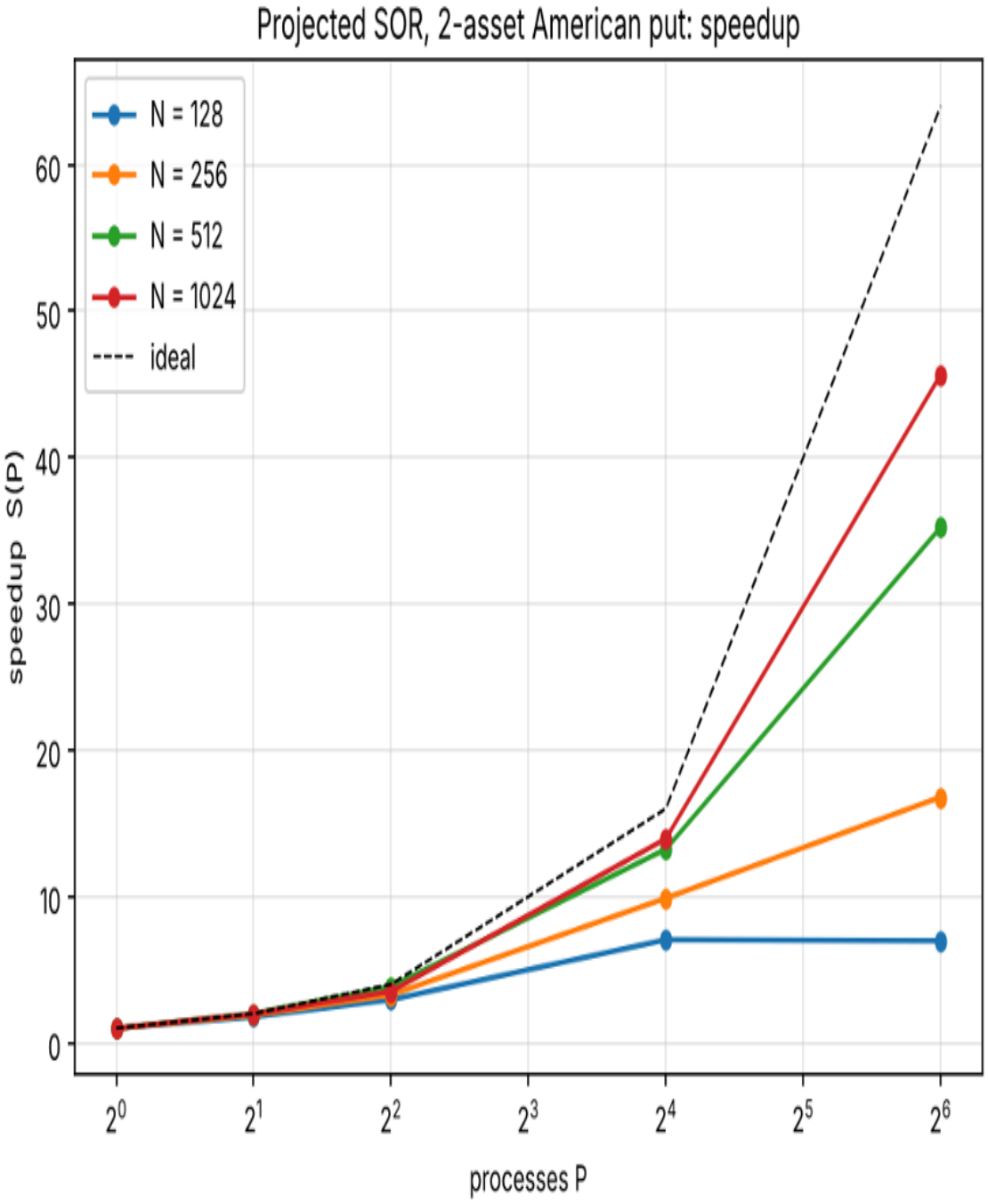


Figure 1. Speedup against core count. The curves fan out strictly by problem size. At M = 128 the curve turns over between P = 16 and P = 64: adding cores there makes the program slower.

4.1 Efficiency is governed by surface-to-volume

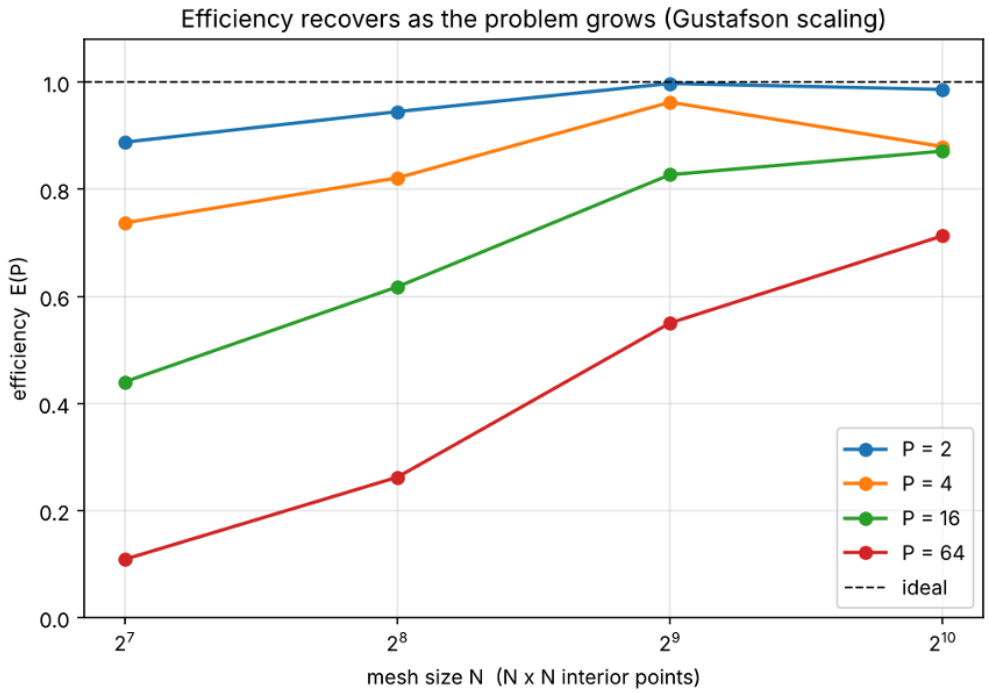


Figure 2. The same measurements re-plotted against mesh size. At every fixed core count, efficiency increases monotonically with the problem. At P = 64 it rises from 10.9% to 71.3%.

Figure 2 is the central performance result and it is a direct illustration of Gustafson's observation [10] that fixed-size scaling understates the value of a parallel machine. The mechanism is not conjectural; it can be read off the decomposition. A rank owning an n×n block performs work proportional to n^2 and communicates a quantity proportional to 4n, so the ratio of communication to computation falls as 1/n. Holding the mesh fixed and increasing P shrinks n and inflates that ratio; growing the mesh restores it.

P	block per rank	halo : interior	measured comm %
1	1024×1024	0.0039	0.3%
2	1024×512	0.006	0.6%
4	512×512	0.0078	3.1%
16	256×256	0.0156	20.1%
64	128×128	0.0312	21.8%

Table 6. The geometric prediction and the measurement, at M = 1024. The halo-to-interior ratio rises by a factor of eight from P = 1 to P = 64 and the measured communication share rises with it.

4.2 Where the method stops paying

At M = 128 and P = 64 each rank holds a 16×16 block bounded by a 64-cell halo, so a quarter of the data a rank owns is boundary. Communication occupies 86.6% of the runtime and the measured speedup falls from 7.05 at P = 16 to 6.98 at P = 64. This is the point at which the decomposition has been carried past its useful limit, and it is more instructive than the rows that scale cleanly. The practical reading is that this solver wants of order 10^4 interior points per rank; below that, the halo dominates.

4.3 Computation scales, communication does not

P	compute (s)	comm (s)	points/rank	updates s $^{-1}$ per rank
1	822.04	2.57	1048576	1.34e+08
2	415.54	2.41	524288	1.33e+08
4	227.21	7.15	262144	1.21e+08
16	47.26	11.89	65536	1.46e+08
64	14.13	3.93	16384	1.22e+08

Table 7. Decomposition of runtime at M = 1024. Per-rank arithmetic throughput varies by less than 20% across a sixty-four-fold change in core count, so essentially all of the efficiency loss is attributable to communication and synchronisation rather than to degraded single-core performance.

The Karp-Flatt metric [11] supports the same conclusion from the opposite direction. At M = 1024 it remains below 0.05 at every core count and falls to 0.0064 at P = 64, whereas at M = 128 it is an order of magnitude larger. A genuinely serial fraction, in the sense of Amdahl [9], would be a property of the algorithm and would not depend on the mesh; that it varies by more than tenfold across problem sizes shows that the overhead being measured is communication, which grows with P and shrinks with N, rather than inherently sequential work. The same inference was reached in Project 2 by a different route.

4.4 A limitation of the instrumentation

At M = 1024 the absolute communication time is not monotone in P: 11.89 s at P = 16 against 3.93 s at P = 64, even though the latter spans three nodes. We decline to propose a mechanism. The reported figure is the slowest rank's own time inside MPI calls, and that quantity conflates genuine message cost with time spent waiting for stragglers. At P = 64 the allocation places 28, 28 and 8 ranks on the three nodes, so the lightly loaded third node finishes early and waits, while the slowest rank, residing on a saturated node, waits least and therefore records a small communication time. Message volume itself decreases strictly with P, since halos shrink, so the observation cannot be explained by traffic alone. Distinguishing the two contributions requires reporting the minimum, mean and maximum communication time across ranks rather than the slowest rank's alone, and we identify this as the natural next measurement rather than advancing an untested explanation here.

4.5 Why no cache cliff appears, and where it would

Project 2 exhibited a pronounced L3 capacity cliff on this same hardware. None appears here, and the reason is algorithmic rather than architectural. In the earlier work each rank held a complete copy of the second operand, so the aggregate footprint per socket scaled as $(P/2)(N^2 + 2N^2/P)\cdot 8$ bytes and therefore grew with the core count until it crossed the 35 MiB shared by the fourteen cores of a socket. A domain decomposition replicates nothing. Each rank holds four arrays over its own block plus a one-cell halo, so the per-socket footprint is approximately $(P/2)\cdot 4(N^2/P)\cdot 8 = 16N^2$ bytes, independent of P. At N = 1024 this is 16.8 MiB, comfortably within the shared cache at every core count, which is consistent with the flat per-rank throughput of Table 7. The analysis yields a falsifiable prediction. Setting $16N^2$ equal to 35 MiB places the crossing at $N \approx 1480$, so a run at N = 2048 on the same 28-core nodes should reproduce a Project-2-style collapse in per-rank throughput while N = 1024 should not. We note also that the monolithic per-socket L3 of the Haswell parts makes them a better instrument for such measurements than a Milan node, whose cache is partitioned per core complex and whose effective capacity therefore depends on rank placement.

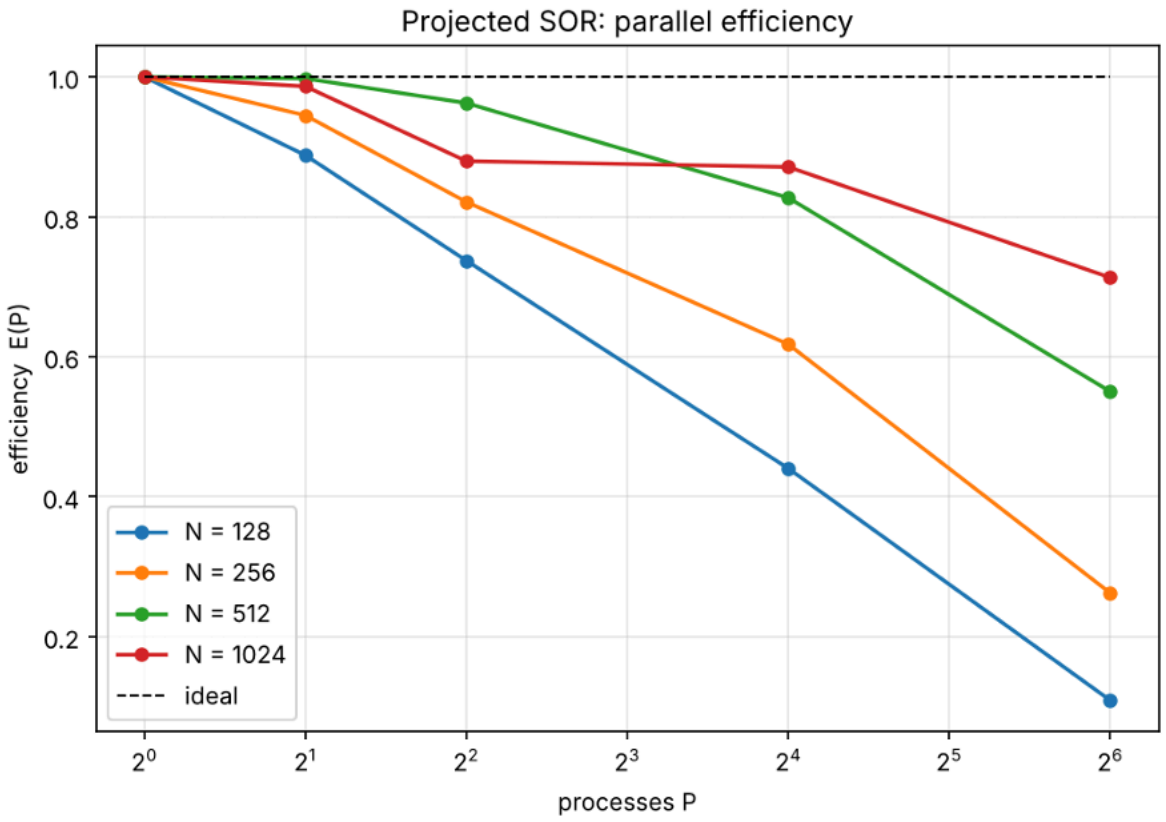


Figure 3. Parallel efficiency against core count.

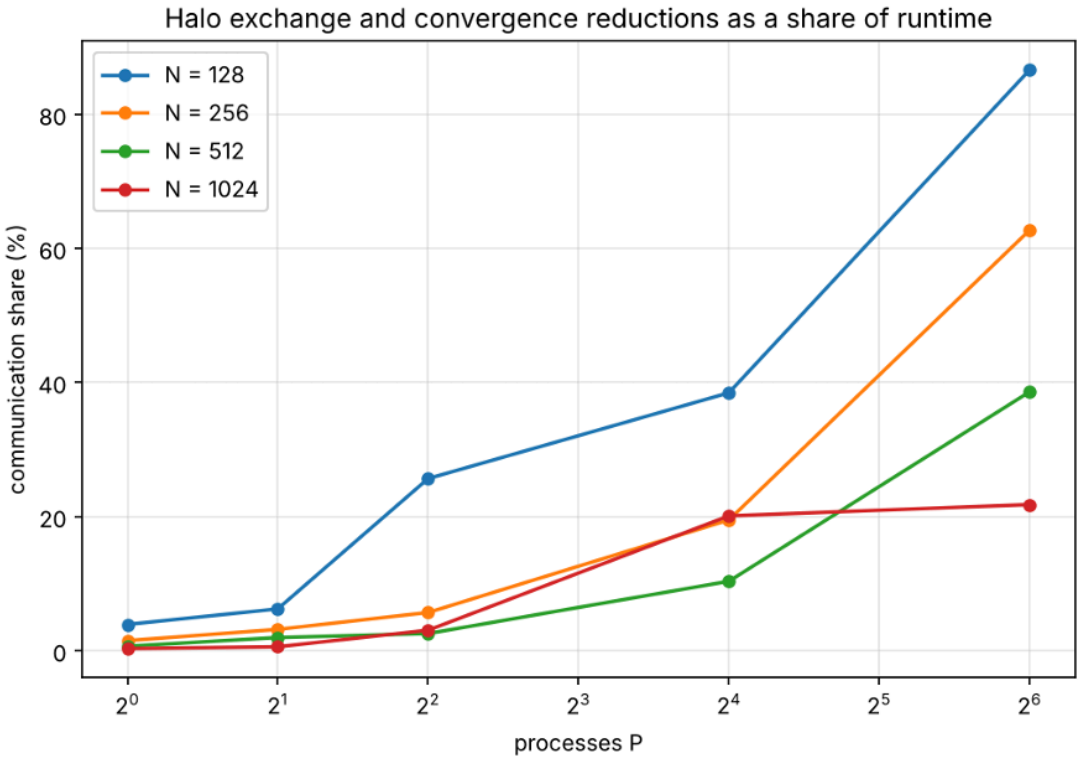


Figure 4. Communication and synchronisation as a share of runtime. The ordering of the curves by problem size is the surface-to-volume law of Section 4.1 in another projection.

5 Conclusions

A two-asset American put has been priced by treating the early-exercise condition as a linear complementarity problem and solving the resulting sequence of sparse systems by projected SOR with red-black ordering, parallelised by two-dimensional domain decomposition. The implementation attains a speedup of 45.65 on 64 cores at a 1024×1024 mesh, at 71.3% efficiency.

Three conclusions seem worth recording. First, the correctness evidence is unusually strong for a problem with no closed-form solution: the solution is invariant under decomposition to ten digits, the European limit reproduces Black-Scholes to within 0.33%, and the deep-in-the-money premium matches $2K(1 - e^{-rT})$ to seven figures. The second and third of these are possible only because the payoff was chosen to be separable and to consist of puts; verification was a design consideration, not an afterthought.

Second, efficiency at fixed core count is governed by the surface-to-volume ratio of the subdomain, and the measured communication share tracks the geometric halo-to-interior ratio across a factor of eight. Practically, the solver requires of order 104 interior points per rank to

remain efficient; below that, as the M = 128 row demonstrates, additional cores make the program slower.

Third, the structural contrast with Project 2 is instructive. Replicated-operand algorithms accumulate memory pressure as they scale and eventually collide with cache capacity; partitioned-data stencil algorithms do not, their per-socket footprint being independent of the core count. This is why no cliff appears in the present measurements, and it furnishes a concrete prediction, $N \approx 1480$, for where one would.

Finally, a remark on the method itself. The entire American feature resides in one pointwise maximum, applied after each relaxation. It consults no neighbor and crosses no subdomain boundary. The free boundary is never located, only revealed. That the analytically intractable component of the problem should also be the component that costs nothing to parallelise is a pleasing accident of the complementarity formulation, and it is the reason projected SOR remains the standard approach to American options on a lattice.

Corrections/Appendix:

Table 2. Disabling the projection must reproduce the sum of two Black-Scholes puts. Relative error falls from 0.437% to 0.325% and then stalls, and the reason is worth stating precisely. The maximum error is nearly independent of M but halves whenever the number of time steps is doubled, so at nt = 50 it is dominated by the first-order accuracy of backward Euler rather than by the mesh. Separately, instrumenting its location places it adjacent to the strike, where the terminal payoff has a kink; that is where the error concentrates spatially, while $\Delta\tau$ sets its magnitude. Crank-Nicolson would address the former and Rannacher start-up [8] the latter. Time steps were held at 50 deliberately, since the performance study requires identical work at every core count and accuracy is not its object.

Section 2.4 The build log records a `-Wstringop-overflow` diagnostic against `MPI_Waitall`; this is a known false positive arising from the zero-length array declaration of `MPI_STATUSES_IGNORE` in the mvapich2 headers, and the code compiles and runs correctly.

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